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ENVIRONMENTAL INFORMATION COLLECTION DEVICE, ENVIRONMENTAL INFORMATION COLLECTION METHOD, AND RECORDING MEDIUM

Abstract

An environmental information collection device acquires user identification information for identifying a user. The environmental information collection device acquires first biometric information of the user. The environmental information collection device authenticates the user by comparing second biometric information associated with the user identification information with the first biometric information acquired by the biometric information acquisition means. The environmental information collection device detects based on the first biometric information that the user is infected with the disease. The environmental information collection device acquires movement history information concerning each area through which the user passed, by using the user identification information. The environmental information collection device outputs a collection request for requesting to collect environmental information concerning a state of each area through which the user passed.

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Background/Summary

TECHNICAL FIELD
[0001] This disclosure relates to a technology for collection information of a region.
BACKGROUND ART
[0002] Currently, artificial satellites are used for various purposes, and are used in some cases to assess disaster and other situations and a warfare condition.
[0003] Moreover, conventionally, systems are known to consider an epidemic state of an infectious disease in each region in a case of providing healthcare services. Patent document 1 describes a system for obtaining movement history information for each individual and determining an infection status of each individual with the infectious disease based on this movement history information. [0004] Patent Document 1: Japanese Patent Application Laid-Open under No. 2022-032426
SUMMARY
[0005] Currently, an imaging plan for satellites to capture images concerning disaster situations and warfare situations is made by collecting information on each area from open sources such as television, magazines, and the Internet. However, because some of the information in open sources may be untrue, it is not always possible to make appropriate imaging plans.
[0006] One object of the present disclosure is to assist in collecting appropriate information of the region.
[0007] According to an example aspect of the present invention, there is provided an environmental information collection device including: [0008] at least one memory configured to store instructions; and [0009] at least one processor configured to execute the instructions to: [0010] acquire user identification information for identifying a user; [0011] acquire first biometric information of the user; [0012] authenticate the user by comparing second biometric information associated with the user identification information with the first biometric information acquired; [0013] detect based on the first biometric information that the user is infected with the disease; [0014] acquire movement history information concerning each area through which the user passed, by using the user identification information; and [0015] output a collection request for requesting to collect environmental information concerning a state of each area through which the user passed.
[0016] According to another example aspect of the present invention, there is provided an environmental information collection method executed by an environmental information collection device, the environmental information collection method comprising: [0017] acquiring user identification information for identifying a user; [0018] acquiring first biometric information of the user; [0019] authenticating the user by comparing second biometric information associated with the user identification information with the first biometric information acquired; [0020] detecting based on the first biometric information that the user is infected with the disease; [0021] acquiring movement history information concerning each area through which the user passed, by using the user identification information; and [0022] outputting a collection request for requesting to collect environmental information concerning a state of each area through which the user passed.

[0023] According to still another example aspect of the present invention, there is provided a non-transitory computer-readable recording medium storing a program causing a computer to execute processing of: [0024] acquiring user identification information for identifying a user; [0025] acquiring first biometric information of the user; [0026] authenticating the user by comparing second biometric information associated with the user identification information with the first biometric information acquired; [0027] detecting based on the first biometric information that the user is infected with the disease; [0028] acquiring movement history information concerning each area through which the user passed, by using the user identification information; and [0029] outputting a collection request for requesting to collect environmental information concerning a state of each area through which the user passed.
Effect
[0030] According to the present disclosure, it is possible to assist in collecting appropriate information of a region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0031] FIG. **1** illustrates an example of a schematic configuration of a health management system.
[0032] FIG. **2** illustrates an example of a hardware configuration of a server.
[0033] FIG. **3** is a diagram schematically illustrating a process by the server.
[0034] FIG. **4** is a block diagram illustrating an example of a functional configuration of the server.
[0035] FIG. **5** illustrates a flowchart of a health management process.
[0036] FIG. **6** illustrates a flowchart of an environmental information collection process.
EXAMPLE EMBODIMENTS
[0037] Preferred example embodiments of the present disclosure will be described with reference to the accompanying drawings.
Example Embodiment
[System Configuration]
[0038] FIG. **1** is an example of a schematic configuration of a health management system **100** to which a server of the present disclosure is applied. The health management system **100** is a system which assists users in managing their health, and provides efficient medical care, by automating counseling to be an initial medical examination by using IDs which identify respective users, and by prioritizing examinations and treatments of patients who have been detected to have a disease.
[0039] Incidentally, the health management system **100** can also be applied as an environmental information collection system which appropriately collects environmental information concerning a state of each region by making highly accurate imaging requests to satellites.
[0040] The user in this disclosure is, for instance, a refugee temporarily staying in a refugee camp or the like. Moreover, the health management system **100** utilizes refugee IDs and cards on which refugee ID are recorded in a refugee ID system operated by the United Nations Organization.
[0041] In the health management system **100**, an input terminal **1**, a server **2**, a satellite system **3**, and a hospital system **4** are communicably connected via a network **5** such as the Internet. The input terminal **1** is an information processing device installed in the refugee camp or the like in each region, and can acquire the refugee ID from the card which a user holds over the input terminal **1**. In addition, the input terminal **1** is provided with a camera and a thermographic camera, and can acquire a face image and a body temperature of the user.
[0042] The server **2** is an information processing device which processes, stores, and exchanges various types of data, analyzes information acquired from the input terminal **1** and the environmental information to be described later to diagnose whether the user is suffering from a predetermined disease, and outputs information for persuading the user to seek a medical

consultation of a doctor if necessary. Here, the user who suffers from a predetermined disease is also referred to as an infected user. In addition, the server **2** sends a request to the satellite system **3** for imaging of each area through which the infected user has passed.

[0043] The satellite system **3** is formed by an information processing device which processes, stores, and exchanges various types of data. The satellite system **3** is a system for managing an operation of an artificial satellite, and records changes by imaging the same point at a specified interval using the artificial satellite and captures images at a predetermined point using the artificial satellite in response to an emergency imaging request. Specifically, if that imaging request is received from the server **2**, the satellite system **3** captures images of an area specified by the artificial satellite, and transmits the captured images or videos as satellite images to the server **2**.

[0044] The hospital system **4** is formed by an information processing device for processing, storing and exchanging various types of data. The hospital system **4** is a system which manages medical appointments and the like of hospitals. Specifically, in a case where the hospital system **4** acquires information such as the refugee ID of each infected user, a symptom, and a disease name from the server **2**, the hospital system **4** makes each medical appointment, and sends each date and time of the medical appointment to the server **2**.

[0045] FIG. **2** is a block diagram illustrating an example of a hardware configuration of the server **2**. As illustrated in FIG. **2**, the server **2** includes an interface (Interface) **11**, a processor **12**, a memory **13**, a recording medium **14**, a display unit **15**, and an input unit **16**. Note that these elements, a user DB **31**, an environment DB **32**, and a report DB **33** are mutually connected via a bus.

[0046] The interface **11** exchanges data with the input terminal **1**, the satellite system **3**, and the hospital system **4**. Moreover, the interface **11** is used by the server **2** to exchange data with a predetermined device connected by wired or wireless communication.

[0047] The processor **12** is a computer such as a CPU (Central Processing Unit) and controls the entire server **2** by executing programs prepared in advance. Incidentally, as the processor **12**, the CPU, a GPU (Graphics Processing Unit), a DSP (Digital Signal Processor), a MPU (Micro Processing Unit), a FPU (Floating Point number Processing Unit), a PPU (Physics Processing Unit), a TPU (Tensor Processing Unit), a quantum processor, a microcontroller, or a combination thereof can be used.

[0048] The memory **13** is formed by a ROM (Read Only Memory) and a RAM (Random Access Memory). The memory **13** stores programs executed by the processor **12**. Moreover, the memory **13** is used as a working memory during executions of various processes performed by the processor **12**.

[0049] The recording medium **14** is a non-volatile and non-transitory recording medium such as a disk-shaped recording medium or a semiconductor memory, and is formed to be detachable from the server **2**. The recording medium **14** records various programs executed by the processor **12**. In a case where the server **2** executes a health management process or an environmental information collection process, corresponding programs recorded in the recording medium **14** are loaded into the memory **13** and executed by the processor **12**.

[0050] The display unit **15** is an LCD (Liquid Crystal Display), for instance, and displays each predetermined image. The input unit **16** is a keyboard, a mouse, a touch panel, or the like, and is used by an operator who manages the server **2**.

[0051] The user DB **31** stores user information in which the refugee ID is associated with a name, biometric information, contact details, movement history information, a diagnosis date and time and, a diagnosis result, and an appointment status. The biometric information is information necessary for facial recognition, for instance, a face image of each user registered in advance. The contact details may be a telephone number, an e-mail address, or the like to provide various notifications to the user. The movement history information is information concerning areas through which the user has passed during the movement, as well as dates and times of a visit of the user to each area. The diagnosis date and time is a date and time in a case where the user is diagnosed whether or not the user is suffering from the disease by analyzing the biometric information obtained from the input terminal **1** and the environmental information described below. The diagnosis result is a result from diagnosing whether or not the user is suffering from the disease. The appointment status is a status of the medical appointment made to the hospital system **4** in a case where the user is infected with the disease, and is information of the hospital name and the date and time of the medical appointment.

[0052] The environment DB **32** stores environmental information concerning the state of each region. Specifically, the environmental information is information concerning the state of each region and the disease related to that state. The state of each region may indicate, for instance, a natural disaster such as a flood, a fire, or a drought, a human-made disaster such as bombing, a landmine, and an epidemic of an infectious disease, a secondary disaster such as a fire and a power outage which are associated with the disaster, or the like. An example of the environmental information is information in which the flood as the state of a region A is associated with a pathogen which is causing an increase in the number of infected in the region A due to the flood. Another example of the environmental information is information in which a fire as the state of a region B is associated with a large number of injuries in the region B due to the fire. Another example of the environmental information is information in which a landmine as a state in a region C is associated with a large number of leg injuries caused by metal fragments in the region C due to the landmine. Another example of environmental information is information in which the epidemic of the infectious disease as a state in a region D is associated with the name of the infectious disease which is prevalent in the region D. Note that the environmental information may be associated with a plurality of states in one area, or may be associated with a plurality of diseases for each state.

[0053] The report DB **33** stores report information in which the refugee ID of the infected user is associated with the symptom and disease name, the movement history information, the state of each area through which the infected user passed, and an outbreak region. The outbreak region is an area where the infected user encountered a state leading to the disease among areas through which the infected user passed. Note that in a case where the outbreak region cannot be specified, it may not be included in the report information. Moreover, the report information may not include movement history information and the state of each area through which the infected user passed, if report information includes the outbreak region and the state of the outbreak region.

[0054] Note that the server **2** of the present disclosure includes the user DB **31**, the environment DB **32**, and the report DB **33** for convenience of explanation, but are not limited to these databases. A type of a DB and a data structure of the DB are arbitrary in a case where necessary data can be obtained. For instance, the user DB **31** may be formed by three DBs: the user DB for storing each refugee ID, the name, the biometric information, and the contact details; a movement history DB for storing the refugee ID and the movement history information; and an appointment DB for storing the diagnosis date and time, the diagnosis result, and the status of the medical appointment.

[0055] FIG. **3** is a diagram schematically illustrating a health management process and an environmental information collection process which are performed by the server **2**. As illustrated in FIG. **3**, the user who is a subject of an initial counseling diagnosis goes to the refugee camp, the hospital, or a government facility where the input terminal **1** is installed, in a case where the user is feeling unwell. Upon arrival, the user holds a card of the user over the input terminal **1** and a face image of the user is captured. Accordingly, the input terminal **1** acquires the refugee ID, the face image, and the body temperature of the user, and transmits the acquired information to the server **2**. The server **2** authenticates the user based on the information acquired from the input terminal **1**, and performs the initial counseling diagnosis in a case where there is no problem.

[0056] In the initial counseling diagnosis, a probability that the user has suffered from a predetermined disease is calculated, and it is determined that the user is infected with the predetermined disease in a case where the probability is equal to or greater than a threshold value. On the other hand, in a case where the probability is less than the threshold value, it is determined that the user is not suffering from the predetermined disease. Specifically, although the initial counseling diagnosis will be described in detail later, in the initial counseling diagnosis, the symptom is specified and the disease name is estimated, based on based on respective determination results of aging of face, the body temperature, and an initial symptom by a symptom AI, and the environmental information of each area through which the user has passed, and the probability that the user is suffering from the predetermined disease is calculated. Note that in a case where there is no environmental information of each area through which the user has passed, the server **2** specifies the symptom and estimate the disease name based on respective determination results of the aging of face, the body temperature, and the initial symptom by the symptom AI, and then calculates the probability of suffering from the predetermined disease. Here, the AI (Artificial Intelligence) is a technique which makes a computer perform intelligent actions such as an inference, a decision-making, and the like which are carried out by humans using intelligence. As will be described later, the symptom AI is a technique that uses machine-learning to determine whether or not initial symptoms of the disease have appeared.

[0057] In a case where the initial-counseling diagnoses that the user has suffered from the predetermined disease, the server **2** updates the report DB **33** by storing information in which the refugee ID of the infected user is associated with the symptom and the disease name, the movement history information, the state of the passing area, and the outbreak region. In addition, in a case where the user is diagnosed as suffering from the predetermined disease by the initial counseling diagnosis, the server **2** makes a medical appointment for the infected user by transmitting information including the refugee ID, the name, and the symptom, the disease name, and the like of the infected user to the hospital system **4**. Next, the server **2** updates the user DB **31** by storing the diagnosis result, the diagnosis date and time, and the status of the medical appointment in the user DB **31** by associating with the refugee ID of the infected user, and reports necessity of counseling by the doctor, the date and time for the medical appointment made to the hospital as a result of the initial counseling diagnosis to the infected user by referring to the contact details. The infected user who has confirmed the report visits the hospital on the date and time of the medical appointment to be examined by the doctor.

[0058] On the other hand, in a case where the initial counseling diagnosis determines that the user is not suffering from the predetermined disease, the server **2** updates the user information by storing the diagnosis result and the diagnosis date and time in the user DB **31** in association with the refugee ID of the user, and reports the user by referring to the contact details that the medical examination by the doctor is unnecessary as a result of the initial counseling diagnosis. In this case, the user does not go to the hospital to be examined by the doctor. Accordingly, by having server **2** perform the initial counseling diagnosis, it is possible to give the infected user a higher priority for the medical consultation by the doctor, thereby improving efficiency of medical care and also preventing a delay of the medical consultation due to a shortage of doctors.

[0059] A report function of the server **2** refers to the report DB **33**, and provides the movement history information of the infected user, the outbreak region, and the like to an imaging plan AI in a case where the infected user has passed through an area to be surveyed by the artificial satellite. The imaging plan AI examines a recommended imaging area based on the provided information, creates an imaging request specifying an area which is preferable to be captured for checking the state of that area, and transmits the created imaging request to the satellite system **3**. If the imaging request is acquired from the server **2**, the satellite system **3** captures the area designated by the artificial satellite, and transmits the satellite images to the server **2**. The server **2** acquires the environmental information by analyzing the acquired satellite images, and updates the environment DB **32** by the acquired environmental information. The environmental information stored in the environment DB **32** are utilized for the initial counseling diagnosis of other users.

[0060] Moreover, the imaging plan AI refers to the user DB **31** based on the movement history information of the infected user and the outbreak region, specifies, as the high-risk user, the user who visited the same area at the same timing as the infected user, and alerts the high-risk user that the high-risk user may have suffered from the same disease as the infected user.
[0061] As described above, the health management system **100** performs the initial counseling diagnosis of the user in consideration of the environmental information, and also acquires new environmental information in response to the imaging request which is based on the result of the initial counseling diagnosis. That is, there is a mechanism in which the environmental information is circulated among the functions of the health management system **100**.
[0062] FIG. **4** is a block diagram illustrating an example of a functional configuration of the server **2**. The server **2** functionally includes an ID acquisition unit **41**, a biometric information acquisition unit **42**, an authentication unit **43**, a movement history acquisition unit **44**, an environmental information acquisition unit **45**, a disease verification unit **46**, an outbreak region specifying unit **47**, an appointment unit **48**, a result report unit **49**, a high-risk user specifying unit **50**, an alert information output unit **51**, and a collection request output unit **52**. The ID acquisition unit **41**, the biometric information acquisition unit **42**, the authentication unit **43**, the movement history acquisition unit **44**, the environmental information acquisition unit **45**, the disease verification unit **46**, the outbreak region specifying unit **47**, the appointment unit **48**, the result report unit **49**, the high-risk user specifying unit **50**, the alert information output unit **51**, and the collection request output unit **52** are realized by the processor **12** executing corresponding programs.
[0063] Note that the report function and a function of the imaging plan AI depicted in FIG. **3** correspond to the outbreak region specifying unit **47**, the appointment unit **48**, the result report unit **49**, the high-risk user specifying unit **50**, the alert information output unit **51**, and the collection request output unit **52** depicted in FIG. **4**.
[0064] The ID acquisition unit **41** acquires the refugee ID of the user to be the subject from the input terminal **1**. The biometric information acquisition unit **42** acquires the face image and the body temperature of the user to be the subject from the input terminal **1**.
[0065] The authentication unit **43** performs the authentication by comparing the face image which is associated in advance with the refugee ID acquired from the input terminal **1**, with the face image acquired from the input terminal **1**. The face image associated with the refugee ID in advance is acquired from the user DB **31** based on, for instance, the refugee ID. Specifically, the authentication unit **43** authenticates the user to be the subject by acquiring features related to facial components such as eyes, a nose, and a mouth from the face image and matching the face images.
[0066] The movement history acquisition unit **44** acquires the movement history information from the user DB **31** based on the refugee ID of the user to be the subject. The movement history information is information concerning each area through which the user passed during the movement and a date and time in a case where the user visited the area. Note that the information indicating the area can be arbitrarily set such as a latitude and longitude, an address, or the like.
[0067] The environmental information acquisition unit **45** acquires the environmental information of the area through which the user has moved from the environment DB **32** based on the movement history information of the user to be the subject. The environmental information is information which matches the state of the region and the disease in the state. Specifically, the environmental information is information in which information specifying the area such as the latitude and longitude, information indicating the state of the region such as the flood, and information concerning the disease such as the name of the disease of numerous outbreaks.
[0068] The disease verification unit **46** performs verification to detect a disease, such as specifying symptoms, calculating a probability of suffering from the disease, and estimating the name of the disease, and the like for the user to be the subject based on the biometric information obtained from the input terminal **1**. In a case where the environmental information acquisition unit **45** acquires the environmental information, the disease verification unit **46** performs the verification based on the

biometric information and the environmental information.

[0069] The disease verification unit **46** includes a facial recognition unit **61**, a body temperature determination unit **62**, and an initial symptom determination unit **63**. The facial recognition unit **61** compares the face image acquired from the input terminal **1** with an aging face image based on a face matching by another service, and determines whether there are any sudden changes in a shape or color of a facial area. The body temperature determination unit **62** determines whether or not the body temperature acquired from the input terminal **1** is abnormal, for instance, whether the body temperature is equal to or more than a threshold value. The initial symptom determination unit **63** is a function by the symptom AI depicted in FIG. **3**, and determines in response to the input of the face image whether or not the initial symptom of the disease has appeared, based on the face image of the user to be the subject by using the machine learning model which has been trained to output whether or not the initial symptom of the disease has appeared on the face.

[0070] The disease verification unit **46** determines that the user to be the subject is unlikely to be suffering from the predetermined disease in a case of no abnormalities in all three viewpoints based on respective results of the facial recognition unit **61**, the body temperature determination unit **62**, and the initial symptom determination unit **63**. At this time, the disease verification unit **46** may calculate the probability of suffering from the disease in a numerical value acquired by a predetermined calculation formula. Note that in a case where the environmental information acquisition unit **45** has acquired the environmental information, the disease verification unit **46** determines the probability that the user to be the subject is infected with the disease, in consideration of the state of the area through which the user to be the subject has passed and the disease in that state. If it is determined that the user is unlikely to have the disease or if the calculated probability is less than a threshold value, the disease verification unit **46** diagnoses that the user to be the subject is not suffering from the predetermined disease, that is, the user is healthy. In a case of diagnosing that the user does not suffer from the predetermined disease, the disease verification unit **46** updates the user DB **31** by storing the refugee ID and the name of the user, a diagnosis result, and a diagnosis date and time.

[0071] On the other hand, in a case of detecting an abnormality in one or more viewpoints based on respective results of the facial recognition unit **61**, the body temperature determination unit **62**, and the initial symptom determination unit **63**, the disease verification unit **46** determines that the probability of the user to be the subject being infected with the predetermined disease is high. Note that in a case where the environmental information acquisition unit **45** has acquired the environmental information, the disease verification unit **46** estimates the disease name in consideration of the state of the area through which the user to be the subject has passed, and the disease in the state. For instance, in a case where it is recognized from the environmental information that Influenza A is prevalent in the region D and the user to be the subject exhibits a symptom of fever, the disease verification unit **46** estimates the disease to be Influenza A. The disease verification unit **46** diagnoses that the user to be the subject has infected with the predetermined disease in a case of determining that the probability that the user is infected with the disease or in a case of determining that the calculated probability is equal to or greater than the threshold value. In other words, the disease verification unit **46** diagnoses that the disease of the user to be the subject is detected.

[0072] In a case of diagnosing that the user to be the subject is infected with the predetermined disease, the disease verification unit **46** updates the report DB **33** by storing information that the refugee ID of the infected user is associated with the symptom and the disease name, the movement history information, the state of the area through which the infected user passed, and the outbreak region.

[0073] The outbreak region specifying unit **47** refers to the report DB **33**, and specifies the area where the infected user encountered a state leading to the disease as the outbreak region. For instance, in a case where the presumed disease name is an enterohemorrhagic *Echerichia coli* infection and the infected user is passing through the region A where the *Echerichia coli* is increasing due to the flood from the report DB **33**, the outbreak region specifying section **47** specifies the region A as the outbreak region. In a case of specifying the outbreak region, the outbreak region specifying unit **47** updates the report DB **33** by further storing information of the outbreak region and the date and time if the infected user visited the outbreak region in the report information corresponding to the infected user.

[0074] In response to the diagnosis by the disease verification unit **46** that the user to be the subject is infected with the predetermined disease, the appointment unit **48** makes a medical appointment for the infected user by transmitting information including the refuge ID and the name of the infected user, the symptom, the disease name, and the like to the hospital system **4**. If the medical appointment is completed, the appointment unit **48** updates the user DB **31** by storing the diagnosis result, the diagnosis date and time, and the status of the medical appointment in association with the refugee ID of the infected user.

[0075] The result report unit **49** reports that the medical examination is not necessary to the user by referring to the contact details if the disease verification unit **46** diagnoses the user to be the subject as healthy. On the other hand, if the user to be the subject is diagnosed by the disease verification unit **46** as suffering from the predetermined disease, the result report unit **49** reports that the medical examination by the doctor is necessary and that the date and time of the medical appointment have been scheduled, the user by referring to the contact details.

[0076] The high-risk user specifying unit **50** refers to the user DB **31** based on the movement history information and the outbreak region of the infected user stored in the report DB **33**, and specifies each user who visited the same area at the same timing as the infected user, to be the high-risk user. The same timing as the infected user is within a predetermined time in reference to a visited date and time of the infected user, and the time is arbitrarily set according to the disease suffered such as one day, one week, one month, or the like.

[0077] The alert information output unit **51** transmits alert information indicating that the disease may be the same as that of the infected user to the contact details of the high-risk user. Thus, even in a case where the symptom has not yet appeared or the symptom is mild, the high-risk user goes to the hospital, explains a situation, and can receive a medical examination. Therefore, it is possible for the high-risk user to be treated at an early stage before the symptom of the high-risk user worsen.

[0078] The collection request output unit **52** refers to the report DB **33**, and if the infected user is passing through the area to be surveyed by the artificial satellite, the collection request output unit **52** examines a recommended imaging area based on the movement history information of the infected user and the outbreak region, creates an imaging request which specifies the area to be captured for checking the state, and transmits the created imaging request to the satellite system **3**. If the imaging request is acquired from the server **2**, the satellite system **3** captures the area designated by the artificial satellite, and transmits the satellite images to the server **2**. The server **2** acquires the environmental information by analyzing the acquired satellite images, and updates the environment DB **32**. Note that the satellite system **3** may report the designated area included in the imaging request and the state of the designated area to government agencies such as ministries and local authorities for use in a disaster management.

[0079] In the configuration described above, the ID acquisition unit **41**, the biometric information acquisition unit **42**, the authentication unit **43**, the movement history acquisition unit **44**, the environmental information acquisition unit **45**, the disease verification unit **46**, the outbreak region specifying unit **47**, the collection request output unit **52**, the facial recognition unit **61**, the body temperature determination unit **62**, and the initial symptom determination unit **63** correspond to respective examples of a user identification information acquisition means, a biometric information acquisition means, an authentication means, a movement history acquisition means, a disease verification means, an outbreak region specifying means, a collection request output means, a facial recognition means, a body temperature determination means, and an initial symptom determination means in this disclosure. Moreover, the report DB **33** included in the server **2** corresponds to an example of a report storage unit.

(Health Management Process)

[0080] Next, a health management process by the server **2** will be described. FIG. **5** is a flowchart of the health management process performed by the server **2**. This health management process is realized by the processor **12** illustrated in FIG. **2** executing a corresponding program prepared in advance.

[0081] If the user who is the subject of the medical consultation is unwell, the user goes to a predetermined administrative facility where the input terminal **1** is installed. If the user arrives at the predetermined administrative facility, the user holds the card of the user over the input terminal **1**, and the face image of the user is captured. Accordingly, the input terminal **1** acquires the refugee ID of the user, the face image, and the body temperature, and transmits the acquired information to the server **2**.

[0082] The server **2** first acquires the refugee ID from the input terminal **1** (step S**101**). Moreover, the server **2** acquires the face image and the body temperature as the biometric information from the input terminal **1** (step S**102**). The server **2** authenticates the user by comparing the face image associated with the refugee ID in advance with the face image acquired from the input terminal **1**. If the user is not a person in question (step S**103**; No) as a result of the authentication, the server **2** terminates the health management process. On the other hand, if the user is the person in question (step S**103**; Yes) as the result of the authentication, the server **2** acquires the movement history information from the user DB **31** based on the refugee ID (step S**104**). Next, the server **2** acquires the environmental information from the environment DB **32** based on the movement history information.

[0083] If there is no environmental information corresponding to the movement history information, that is, the environmental information of each area through which the user has passed is not stored in the environment DB **32** (step S**105**; No), the server **2** verifies the disease of the user based on the biometric information, and calculates the possibility that the user is infected with the disease (step S**108**). On the other hand, if there is the environmental information corresponding to the movement history information (step S**105**; Yes), the server **2** acquires the environmental information of each area through which the user has passed from the environment DB **32** (step S**106**). The server **2** verifies the disease of the user based on the acquired environmental information and the biometric information, and calculates the possibility that the user is infected with the disease (step S**107**). If the probability of suffering from the disease is calculated, the server **2** determine whether the probability is equal to or greater than a threshold value (step S**109**).

[0084] If the probability of suffering from the disease is less than the threshold value (step S**109**; No), the server **2** diagnoses the user as healthy, and goes to a process of step S**111**. On the other hand, if the probability of suffering from the predetermined disease is equal to or greater than the threshold value (step S**109**; Yes), the server **2** diagnoses that the user is suffering from the disease, and makes a medical appointment in cooperation with the hospital system **4** (step S**110**). Next, the server **2** reports a diagnostic result to the user (step S**111**). Specifically, the server **2** reports, as a result, the user diagnosed as healthy, that the medical consultation by the doctor is unnecessary. On the other hand, the server **2** reports, as the result, the user diagnosed as suffering from the predetermined disease that the medical consultation by the doctor is necessary and the date and time of the medical appointment made to the hospital.

[0085] Moreover, if the user is diagnosed as suffering from the predetermined disease, the server **2** updates the report DB **33** by storing information in which the refugee ID of the infected user is associated with the symptom and the disease name, the movement history information, and the outbreak region, and the state of the outbreak region (step S**112**).

[0086] Next, the server **2** refers to the user DB **31** based on the movement history information of the infected user and the outbreak region stored in the report DB **33**, specifies, as the high-risk user, the user who visited the same area at the same timing as the infected user, and alerts the user (step S**113**). Thus, the health management process is terminated.

[0087] Note that in the present disclosure, in a case where the user is diagnosed as suffering from the predetermined disease, the server **2** makes the medical appointment in cooperation with the hospital system **4**, but is not limited thereto. Instead of making the medical appointment, the server **2** may simply report the user of the result that the medical consultation by the doctor is necessary, thereby prompting the user to seek a medical consultation of the doctor. Moreover, the server **2** may send information concerning the symptom of the user and the estimated disease name to the hospital system **4** as a counseling report, and propose a preferential medical consultation of the infected user to the doctor, or repot a treatment proposal to the doctor.

[0088] Accordingly, the server **2** assists the infected user to receive the medical consultation preferentially based on the state of each of areas and the symptoms of users who visited the areas. By this assistant, it is possible to improve the efficiency of medical consultations by doctors. Moreover, based on the state of each area and an actual condition of the disease by which each user is infected in that state, it is possible to alert the user with a history of visiting that area and prompt the user to seek a medical consultation from the doctor at an appropriate timing. Therefore, it is possible for the server **2** to improve efficiency of each medical service and provide treatment tailored to the needs of each individual.

(Environmental Information Collection Process)

[0089] Next, the environmental information collection process by the server **2** will be described. FIG. **6** is a flowchart of the environmental information collection process performed by the server **2**. This environmental information collection process is realized by executing a corresponding program prepared in advance by the processor **12** illustrated in FIG. **2**.

[0090] If the infected user passes the area be surveyed by the artificial satellite, the server **2** refers to the report DB **33** and examines the recommended imaging area based on the movement history information of the infected user and the outbreak region (S**201**). Next, the server **2** creates an imaging request which specifies the area to be captured for checking the state, and transmits the request to the satellite system **3** (step S**202**). The satellite system **3** captures images of the area designated by the artificial satellite and transmits satellite images to the server **2**. The server **2** acquires the environmental information by analyzing the satellite images acquired from the satellite system **3**, and updates the environment DB **32** (step S**203**). After that, the environmental information collection process is terminated.

[0091] In the manner described above, it is possible for the server **2** to accurately understand what information is desired to be acquired next based on previous satellite images and information concerning the health care of the user, and to make an appropriate imaging request which specifies a next imaging location by the artificial satellite. In other words, it is possible to predict a damage situation from the state of each area and the symptom of the user who visited that area, and it is possible to send the appropriate imaging request to the satellite system **3**.

[0092] Moreover, it is possible for the server **2** to make the imaging request in consideration of quantitative information concerning the user who have actually visited the area acquired by the health management system **100**, while utilizing information from open sources such as a television, a magazine, the Internet, and the like. Therefore, it is possible to realize the imaging request with higher accuracy.

First Modification

[0093] The state of the region area not limited to the disaster, but may be a state of crime. For instance, it is assumed that a region E has a high crime rate, and that a power line is artificially disconnected due to a robbery and the like, resulting in a power outage. In this case, the environmental information is, for instance, information which associates the state in area E such as a man-made power outage, with the fact that there are a large number of injuries due caused by assault and injury in the region E as a result of the power outage.

Second Modification

[0094] In the example embodiment described above, the server **2** authenticates each user using the face image acquired from the input terminal **1**, but the present disclosure is not limited thereto. Any method such as a fingerprint authentication, a vein authentication, a voice authentication, and an iris authentication can be applied if an authentication method is performed based on the biometric information which the server **2** acquires from the input terminal **1**.

Third Modification

[0095] The result of the initial counseling diagnosis and alert information are sent to the user in accordance with the contact details; however, in a case where the user does not have a smartphone or the like and there are no contact details, the result and the alert information may be sent to the input terminal **1**, or be presented to the user by a staff or the like at the refugee camp receiving by a predetermined terminal and posting the result and the alert information.

Fourth Modification

[0096] In the example embodiments described above, the server **2** specifies the outbreak region or recognizes the conditions of the outbreak region, by referring to each DB, and creates the imaging request, but the processes of the server **2** are not limited thereto, and the server **2** may create the imaging request using the machine learning model. In this case, the server **2** creates the imaging request according to the symptom and the movement history information of the infected user by using the machine learning model which is trained to output an optimized imaging request in accordance with the symptom and the movement history information of each user. After that, the server **2** transmits the imaging request output from the machine learning model to the satellite system **3**.

[0097] A part or all of the example embodiments (including the modifications, the same hereinafter) described above may also be described as the following supplementary notes, but not limited thereto.

(Supplementary Note 1)

[0098] An environmental information collection device comprising: [0099] a user identification information acquisition means configured to acquire user identification information for identifying a user; [0100] a biometric information acquisition means configured to acquire first biometric information of the user; [0101] an authentication means configured to authenticate the user by comparing second biometric information associated with the user identification information with the first biometric information acquired by the biometric information acquisition means; [0102] a disease verification means configured to detect based on the first biometric information that the user is infected with the disease; [0103] a movement history acquisition means configured to acquire movement history information concerning each area through which the user passed, by using the user identification information; and [0104] a collection request output means configured to output a collection request for requesting to collect environmental information concerning a state of each area through which the user passed.

(Supplementary Note 2)

[0105] The environmental information collection device according to supplementary note 1, wherein [0106] the environmental information collection device is connected to a satellite system, and [0107] the collection request output means sends a request for capturing an image of each area through which the user passed, to the satellite system.

(Supplementary Note 3)

[0108] The environmental information collection device according to supplementary note 1, wherein the environmental information is information of a state of each area and disease corresponding to the state of each area.

(Supplementary Note 4)

[0109] The environmental information collection device according to supplementary note 1,

wherein [0110] the biometric information includes a face image and a body temperature of the user; and [0111] the disease verification means specifies a symptom of the user based on the face image and the body temperature, calculates a probability that the user is infected with the disease, and detect presence of the disease in the user in a case where the probability is equal to or greater than a threshold value.
(Supplementary Note 5)
[0112] The environmental information collection device according to supplementary note 4, further comprising a report storage unit configured to store report information in which the user identification information of an infected user for whom the presence of the disease is detected is associated with the symptom and the movement history information, [0113] wherein the processor outputs the collection request based on the report information.
(Supplementary Note 6)
[0114] The environmental information collection device according to supplementary note 5, further comprising an outbreak region specifying means configured to specify, based on the report information, an area where the infected user encountered a state leading to the disease as an outbreak region, [0115] wherein the collection request output means outputs the collection request concerning the outbreak region.
(Supplementary Note 7)
[0116] The environmental information collection device according to supplementary note 4, wherein the disease verification means includes [0117] a facial recognition means configured to determine whether or not one or more of a shape and color of the facial area is abnormal, [0118] a body temperature determination means configured to determine whether or not a body temperature of the user is equal to or greater than a threshold value, and [0119] an initial symptom determination means configured to determine whether or not an initial symptom has occurred based on the face image of the user by using a machine learning model trained to output whether or not the initial symptom of the disease appearing on a face has occurred, in response to an input of the face image, [0120] wherein the disease verification means specifies the symptom of the user based on respective results of the facial recognition means, the body temperature determination means, and the initial symptom determination means, and calculates the probability that the user is infected with the disease.
(Supplementary Note 8)
[0121] The Environmental Information Collection Device According to supplementary note 4, wherein the collection request output means outputs a collection request corresponding the symptom and a movement history of the user by using a machine learning model trained to output an optimized collection request in response to the symptom and the movement history of the user.
(Supplementary Note 9)
[0122] An environmental information collection method executed by an environmental information collection device, the environmental information collection method comprising: [0123] acquiring user identification information for identifying a user; [0124] acquiring first biometric information of the user; [0125] authenticating the user by comparing second biometric information associated with the user identification information with the first biometric information acquired; [0126] detecting based on the first biometric information that the user is infected with the disease; [0127] acquiring movement history information concerning each area through which the user passed, by using the user identification information; and [0128] outputting a collection request for requesting to collect environmental information concerning a state of each area through which the user passed.
(Supplementary Note 10)
[0129] A program causing a computer to execute processing of: [0130] acquiring user identification information for identifying a user; [0131] acquiring first biometric information of the user; [0132] authenticating the user by comparing second biometric information associated with the user identification information with the first biometric information acquired; [0133] detecting based on

the first biometric information that the user is infected with the disease; [0134] acquiring movement history information concerning each area through which the user passed, by using the user identification information; and [0135] outputting a collection request for requesting to collect environmental information concerning a state of each area through which the user passed.
[0136] While the present disclosure has been described with reference to the example embodiments and examples, the present disclosure is not limited to the above example embodiments and examples. Various changes which can be understood by those skilled in the art within the scope of the present disclosure can be made in the configuration and details of the present disclosure.
[0137] This application is based upon and claims the benefit of priority from Japanese Patent Application 2024-035055, filed on Mar. 7, 2024, the disclosure of which is incorporated herein in its entirety by reference.
DESCRIPTION OF SYMBOLS
[0138] **1** Input terminal [0139] **2** Server [0140] **3** Satellite system [0141] **4** Hospital system [0142] **5** Network [0143] **11** Interface [0144] **12** Processor [0145] **13** Memory [0146] **14** Recording medium [0147] **15** Display unit [0148] **16** Input unit [0149] **31** User DB [0150] **32** Environment DB [0151] **33** Report DB [0152] **100** Health management system

Claims

1. An environmental information collection device comprising: at least one memory configured to store instructions; and at least one processor configured to execute the instructions to: acquire user identification information for identifying a user; acquire first biometric information of the user; authenticate the user by comparing second biometric information associated with the user identification information with the first biometric information acquired; detect based on the first biometric information that the user is infected with the disease; acquire movement history information concerning each area through which the user passed, by using the user identification information; and output a collection request for requesting to collect environmental information concerning a state of each area through which the user passed.
2. The environmental information collection device according to claim 1, wherein the environmental information collection device is connected to a satellite system, and the processor sends a request for capturing an image of each area through which the user passed, to the satellite system.
3. The environmental information collection device according to claim 1, wherein the environmental information is information of a state of each area and disease corresponding to the state of each area.
4. The environmental information collection device according to claim 1, wherein the biometric information includes a face image and a body temperature of the user; and the processor specifies a symptom of the user based on the face image and the body temperature, calculates a probability that the user is infected with the disease, and detect presence of the disease in the user in a case where the probability is equal to or greater than a threshold value.
5. The environmental information collection device according to claim 4, further comprising a report storage unit configured to store report information in which the user identification information of an infected user for whom the presence of the disease is detected is associated with the symptom and the movement history information, wherein the processor outputs the collection request based on the report information.
6. The environmental information collection device according to claim 5, wherein the processor is further configured to specify, based on the report information, an area where the infected user encountered a state leading to the disease as an outbreak region, wherein the processor outputs the collection request concerning the outbreak region.
7. The environmental information collection device according to claim 4, wherein in a case of

detecting based on the first biometric information that the user is infected with the disease, the processor performs to determine whether or not one or more of a shape and color of the facial area is abnormal, determine whether or not a body temperature of the user is equal to or greater than a threshold value, and determine whether or not an initial symptom has occurred based on the face image of the user by using a machine learning model trained to output whether or not the initial symptom of the disease appearing on a face has occurred, in response to an input of the face image, wherein the processor specifies the symptom of the user based on respective results from determining whether or not one or more of the shape and the color of the facial area is abnormal, determining whether or not the body temperature of the user is equal to or greater than a threshold value, and determining whether or not the initial symptom has occurred, and calculates the probability that the user is infected with the disease.

8. The environmental information collection device according to claim 4, wherein the processor outputs a collection request corresponding the symptom and a movement history of the user by using a machine learning model trained to output an optimized collection request in response to the symptom and the movement history of the user.

9. An environmental information collection method executed by an environmental information collection device, the environmental information collection method comprising: acquiring user identification information for identifying a user; acquiring first biometric information of the user; authenticating the user by comparing second biometric information associated with the user identification information with the first biometric information acquired; detecting based on the first biometric information that the user is infected with the disease; acquiring movement history information concerning each area through which the user passed, by using the user identification information; and outputting a collection request for requesting to collect environmental information concerning a state of each area through which the user passed.

10. A non-transitory computer-readable recording medium storing a program causing a computer to execute processing of: acquiring user identification information for identifying a user; acquiring first biometric information of the user; authenticating the user by comparing second biometric information associated with the user identification information with the first biometric information acquired; detecting based on the first biometric information that the user is infected with the disease; acquiring movement history information concerning each area through which the user passed, by using the user identification information; and outputting a collection request for requesting to collect environmental information concerning a state of each area through which the user passed.